BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

CHEMISTRY 9th — ANNUAL EXAM 2021 — SECTION –A

Time Allowed: 20 minutes | **Maximum Marks:** 12

Note: Q1: Choose the correct Answer.

Q1. Modern periodic table is based on .....

(A) Mass number
(B) Nucleon's
(C) Atomic number
(D) Neutrons

Q2. Shielding effect is due to .....

(A) Neutrons
(B) Protons
(C) Nucleons
(D) Electrons

Q3. There is ..... covalent bond in nitrogen molecule (N_2)

(A) Single
(B) Double
(C) triple
(D) Additive

Q4. A caution with a charge is called .....

(A) An electron
(B) A molecule
(C) A metal
(D) An ion

Q5. Which one of the following is a crystalline solid?

(A) Sodium chloride
(B) Charcoal
(C) Candle wax
(D) Rubber

Q6. Alloy is an example ..... solution.

(A) Solid in solid
(B) Solid in liquid
(C) Solid in gas
(D) Liquid in solid

Q7. The branch of chemistry which deals with inter conversion of electrical energy and mechanical energy is called .....

(A) Physical chemistry
(B) Analytical Chemistry
(C) Electro chemistry
(D) Industrial chemistry

Q8. The oxide of calcium, CaO is .....

(A) Acidic
(B) Basic
(C) Ampoteric
(D) Neutral

Q9. A substance consists of two or more substances chemically combined in fixed ratio by mass is called .....

(A) Element
(B) Compound
(C) Mixture
(D) Solution

Q10. One Avogadro's number contains ..... particles.

(A) 6.023×10^{23}
(B) 6.023×10^{19}
(C) 6.023×10^{23}
(D) 6.023×10^{31}

Q11. The maximum number of electrons in a particular orbit is given by the formula .....

(A) $2n$
(B) $2n^2$
(C) $2n^2-2$
(D) $2n^2+2$

Q12. L-Shell has sub shell (s)

(A) S
(B) S,P
(C) S,p,d
(D) S,p,d,

SECTION –B

Maximum Marks: 32

Q1. (Marks 32) Q1: Attempt any eight of the following. All carry equal marks.

i: Differentiate between gram atomic mass and gram molecular mass with example.

ii: How many moles are there in 8.9×10^{23} hydrogen atom?

iii: Write down any four points of Rutherford atomic model?

iv: Draw electronic configuration of (a) Ne^{10}, (b) Cl^{17}

v: What is shielding effect? Give its trends in periodic table.

vi: Make a general sketch of the periodic table without showing s,p, d and f-block elements.

vii: Explain coordinate covalent bond with at least one example?

viii: Diamond is crystalline allotropic form of carbon, example.

ix: Why we stir paints before use it?

x: What is electroplating? Give two purposes of electroplating.

xi: Write at least four uses of sodium.

SECTION -C

Maximum Marks: 21

Note: Note: Attempt any Three questions.

Q2.

a: What is chemistry? Discuss the contribution of Jabber-ibn-Hayan in the field of chemistry.

b: Why Dalton atomic theory is considered as a base for modern atomic concepts?

Q3.

a: Discuss ionization potential. Give its trends in periodic table.

b: Explain hydrogen bonding support your answer with suitable example.

Q4.

a: Define vapour pressure. What are the factors affecting the vapour pressure?

b: Calculate the morality of 5.00 cm^3 of solution containing 7.50 grams of CH_3OH.

Q5.

a: Discuss the preparation of sodium hydroxide (NaOH) from brine along with its important reactions.

b: Give the reactions of magnesium (Mg) with (i) HCl (ii) Cl_2 (iii) O_2